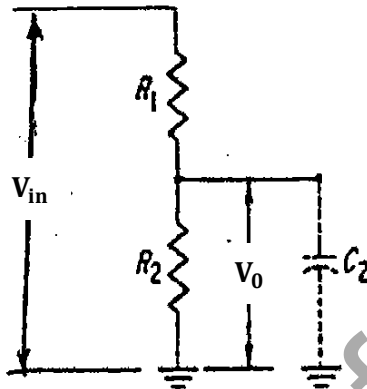


Sheet 3: Attenuators

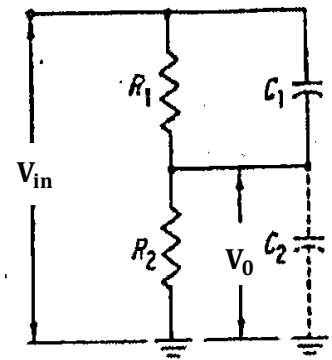
- 1) A voltage divider is used as attenuator in an instrument. The resistances R_1 and R_2 are of $90\text{ k}\Omega$ and $10\text{ k}\Omega$. Find the voltage gain and attenuation in dB. If the transistor and stray cap C_2 across the $10\text{ k}\Omega$ is of 10 pF . Find the value of C_1 across $90\text{ k}\Omega$ resistance that compensate for C_2 .



(a) Uncompensated circuit

Solution:

- $A = \frac{V_o}{V_{in}} = \frac{R_2}{R_2 + R_1} = \frac{10}{10 + 90} = 0.1$
- $a = \frac{V_{in}}{V_o} = \frac{1}{A} = 10$
- $a \text{ in dB} = 20 \log a = 20 \log 10 = 20 \text{ dB}$



(b) Compensated circuit

- $X_C = \frac{1}{2\pi fC}$
- Low freq. $f \approx 0 \Rightarrow X_C = \frac{1}{2\pi fC} = \frac{1}{0} = \infty \Rightarrow A = \frac{V_o}{V_{in}} = \frac{R_2}{R_2 + R_1}$
- High freq. $f \approx \infty \Rightarrow X_C = \frac{1}{2\pi fC} = \frac{1}{\infty} = 0 \Rightarrow A = \frac{V_o}{V_{in}} = \frac{0}{V_{in}} = 0$
- $\frac{R_1}{R_2} = \frac{X_{C1}}{X_{C2}} = \frac{2\pi fC_2}{2\pi fC_1} = \frac{C_2}{C_1} \Rightarrow R_1 C_1 = R_2 C_2$
- $C_1 = \frac{R_2 C_2}{R_1} = \frac{10 \times 10^3 \times 10 \times 10^{-9}}{90 \times 10^3} = \frac{10}{9} \text{ pF}$

2) Find the characteristic resistance of the symmetrical T- attenuator shown in Fig. 1.

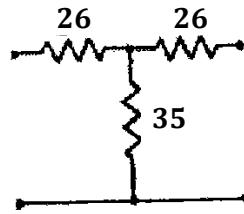
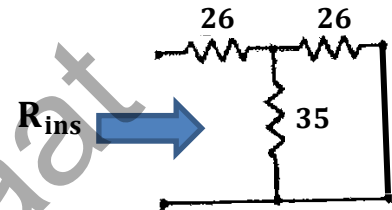


Fig. 1

Solution:

➤ $R_o = \sqrt{R_{ins} * R_{ino}}$

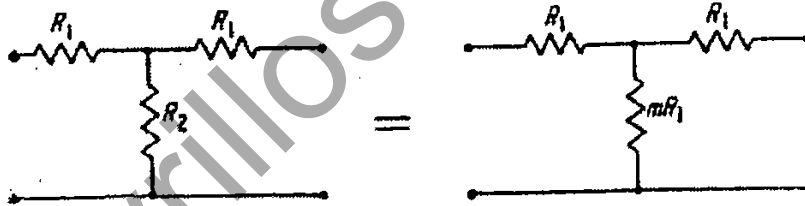
➤ Form Fig. 1, $R_{ino} = 26 + 35 = 61 \Omega$



➤ $R_{ins} = 26 + (35 // 26) = 26 + \frac{35 * 26}{35 + 26} = 26 + 15 = 41 \Omega$

➤ $R_o = \sqrt{R_{ins} * R_{ino}} = \sqrt{41 * 61} = 50 \Omega$

or:



➤ $m = \frac{R_2}{R_1} = \frac{35}{26} = 1.34615$

➤ $R_o = R_1 \sqrt{1 + 2m} = 26 \sqrt{1 + 2 * 1.34615} = 50 \Omega$

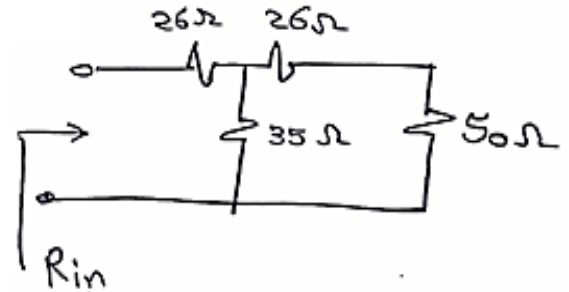
➤ $a = \frac{V_{in}}{V_o} = \frac{1 + m + \sqrt{1 + 2m}}{m} = \frac{1 + 1.346 + \sqrt{1 + 2 * 1.346}}{1.346} = 3.17$

➤ $a \text{ in dB} = 20 \log a = 20 \log 3.17 = 10 \text{ dB}$

- 3) Prove that the input resistance of attenuator of Fig. 1 is $50\ \Omega$ when loaded by its characteristic resistance.

Solution:

$$\begin{aligned} \text{➤ } R_{ins} = R_{th} &= [(26 + 50) // 35] + 26 \\ &= [76 // 35] + 26 = 24 + 26 = 50\ \Omega = R_o \end{aligned}$$



- 4) Design a 70 dB, 500 Ω attenuator using 3 basic T- sections.

Solution:

$$\text{➤ } 70\text{ dB} = 20\text{ dB} + 20\text{ dB} + 30\text{ dB}$$

➤ For two section with 20 dB, 500 Ω :

$$\text{➤ } 20\text{ dB} = 20 \log a \Rightarrow a = 10$$

$$\text{➤ } R_1 = \frac{a-1}{a+1} R_o = \frac{10-1}{10+1} * 500 = 409.1\ \Omega$$

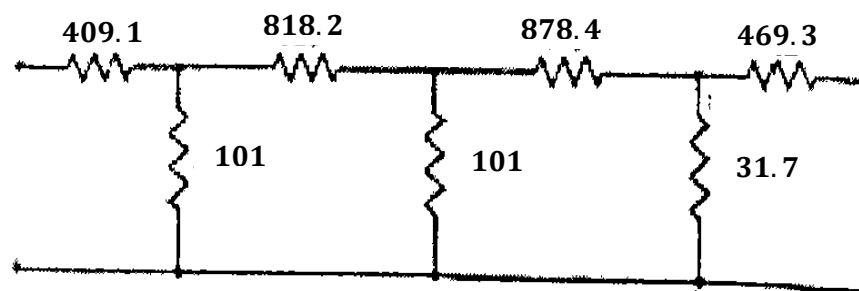
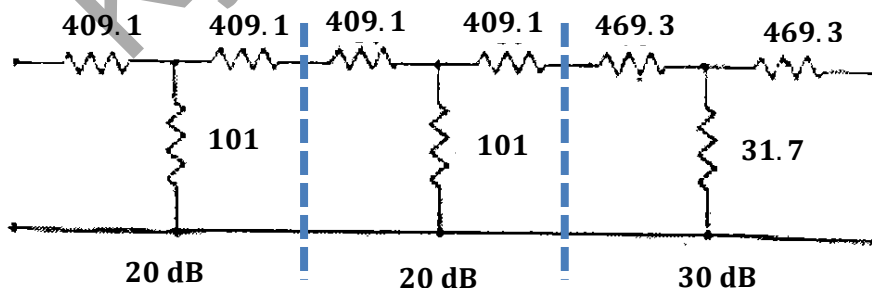
$$\text{➤ } R_2 = \frac{2a}{a^2-1} R_o = \frac{2*10}{10^2-1} * 500 = 101\ \Omega$$

➤ For one section with 30 dB, 500 Ω :

$$\text{➤ } 30\text{ dB} = 20 \log a \Rightarrow a = 31.62$$

$$\text{➤ } R_1 = \frac{a-1}{a+1} R_o = \frac{31.62-1}{31.62+1} * 500 = 469.3\ \Omega$$

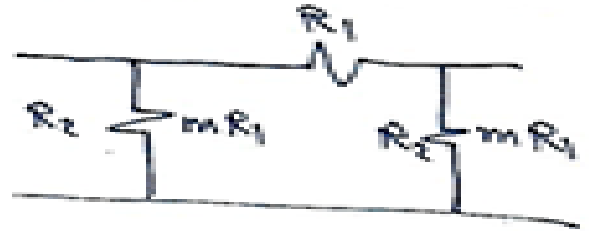
$$\text{➤ } R_2 = \frac{2a}{a^2-1} R_o = \frac{2*10}{31.62^2-1} * 500 = 31.7\ \Omega$$



5) Design a π -attenuator with 20 dB, 50 Ω .

Solution:

- $20 \text{ dB} = 20 \log a \Rightarrow a = 10$
- $R_1 = \frac{a^2 - 1}{2a} R_o = \frac{10^2 - 1}{2 \cdot 10} * 50 = 247.5 \Omega$
- $R_2 = \frac{a + 1}{a - 1} R_o = \frac{10 + 1}{10 - 1} * 50 = 61.1 \Omega$



6) A signal source with a source resistance 200 Ω is to be used in 50 Ω , 4 dB T- section, what is the value of padded source resistance?

Solution:

- $4 \text{ dB} = 20 \log a \Rightarrow a = 1.585$
- $k = \frac{R_s}{R_o} = \frac{200}{50} = 4$

$$R_{\text{out}}(\text{Thevenin Resistance}) = R_o \left[1 + \frac{2}{a^2 \left(\frac{k+1}{k-1} \right) - 1} \right]$$

$$= 50 \left[1 + \frac{2}{(1.585)^2 \left(\frac{4+1}{4-1} \right) - 1} \right] = 81.377 \Omega$$

